Binance Gold & Silver Perpetuals and ETF Markets

Event study: impact on volatility, liquidity, and integration

KSE Python course project

> **Speaker notes:** Introduce the project in one sentence: we ask whether the launch of Binance TRADFI gold and silver perpetuals changed how the established ETF markets (GLD, SLV) behave. Mention event date: 2026-01-05.

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1. Research question

Did the introduction of Binance TRADFI gold and silver perpetual futures (XAUUSDT, XAGUSDT) change:

- **Volatility** of gold and silver ETFs (GLD, SLV)
- **Liquidity** (volume, Amihud illiquidity)
- **Price integration** (correlation ETF–perpetual): we *have* data **after** the launch (rolling correlation GLD–XAU, SLV–XAG); we do **not** have pre-event integration (Binance did not exist), so we describe post-event integration only, not a pre/post test.

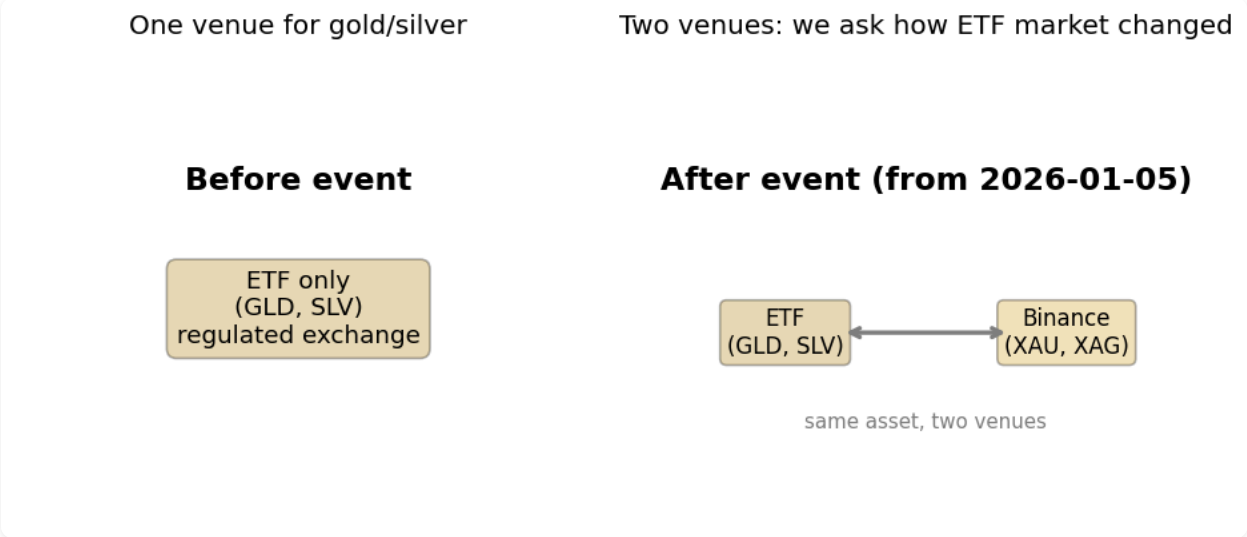
Event: launch date **2026-01-05**.

> **Speaker notes:** State the research question. For integration, clarify: we have correlation data only post-event; we describe how strong integration is after the launch, but we cannot test "integration increased" because there is no pre-period for Binance.

2. Why this story matters

Gold and silver are traded as **ETFs** (GLD, SLV) on regulated exchanges. In early 2026, Binance started listing **perpetuals** on the same assets (XAUUSDT, XAGUSDT). So now there are **two places** to get exposure to the same thing.

We ask: did the **ETF market change** after that? More volatile? More trading? More linked to the new venue? This matters for traders, market-makers, and regulators.



> *"The introduction of derivatives can change spot markets—volatility, liquidity, and how prices move together."*
> Augustin, Rubtsov & Shin (2023), Management Science — Bitcoin futures and spot markets.

> **Speaker notes:** Tell the story: two venues, same asset. Our question is not only "did volatility go up?" but "how did the structure of trading change?"

3. Data: sources and periods

Source	What we get	Frequency	Rows	Note
Binance API	XAUUSDT, XAGUSDT (OHLCV), funding, OI (TRADIFI_PERPETUAL)	1m	67k+	From 2026-01-05 (XAU), 2026-01-07 (XAG)
Yahoo Finance	ETFs: GLD, SLV, IAU, GDX, SIL. Controls: DXY, VIX, TNX. GLD/SLV 1m	Daily; 1m last 7d	118 daily; 1m limited	1m only last ~7 days
FRED	DGS10 — 10Y Treasury rate (macro control)	Daily	in daily panel	Same role as TNX, from Fed data

Panels: Daily = 118 rows (event study, OLS). 1-minute = 67k+ rows (intraday analysis).
Integration: post-event only (rolling corr, beta; no Binance before 2026-01-05).

DXY = dollar index; *VIX* = volatility index; *TNX* / *DGS10* = 10Y yield (Yahoo / FRED). Used as controls.

> **Speaker notes:** One table, row counts clear. Panels and integration in one line. FRED

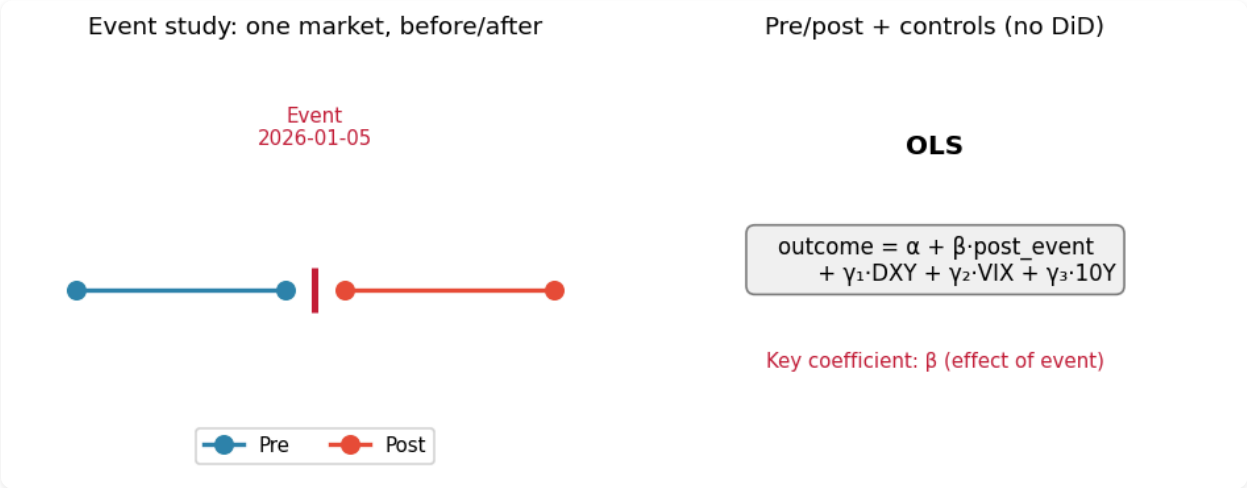
> DGS10 = 10Y rate from Fed.

4. How we use the data

- **Daily panel** (features_daily.csv): 118 days × 87 variables. Used for event study—pre/post means, t-tests, OLS with `post_event` and controls (DXY, VIX, 10Y).
- **1-minute panel** (features_1m.csv): one row per Binance XAUUSDT 1m candle, daily ETF/controls merged by date. We use it for intraday return distribution and daily realized vol from 1m.
- **Minute-level correlation**: last 7 days where Yahoo 1m (GLD/SLV) overlaps with Binance 1m—we compute same-frequency correlation GLD–XAU, SLV–XAG (another integration metric, post-event only).
- **Integration (descriptive)**: rolling correlation and beta in the daily panel and in the key-results table; we report post-event level and evolution only (no pre/post test for integration).

> **Speaker notes:** Daily for regressions and tests; 1m for scale and minute correlation. Integration: we use the data to describe how strong the link is after the launch, not to test a change.

5. Event and method



Design: Event study with pre/post comparison. No control group → we do **not** use difference-in-differences.

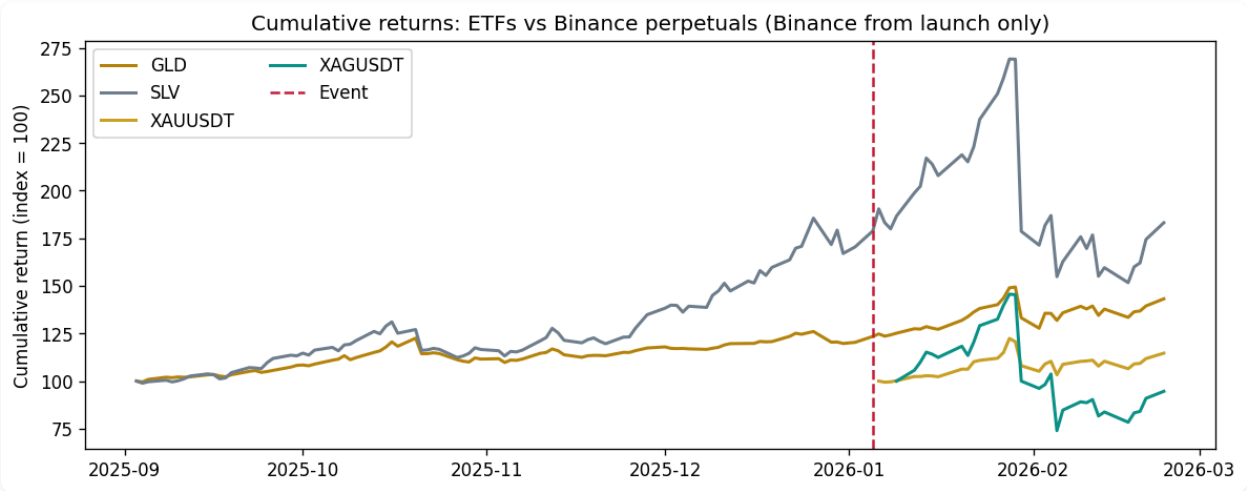
Tests: (1) Welch t-tests (pre vs post); (2) OLS with outcome = volatility or liquidity, regressor = `post_event` + **controls (DXY, VIX, 10Y)**. HAC (Newey–West) SE; winsorization 5%/95%.

Why controls? DXY (dollar), VIX (volatility index), 10Y yield (Yahoo TNX / FRED DGS10) capture general market conditions. By including them in OLS, the coefficient on `post_event` measures the effect of **the listing itself**, not just "the market moved"—so we can argue that the change in ETF volatility/volume is due to the Binance launch, not confounded by macro.

Literature: Augustin et al. (2023), Management Science — impact of Bitcoin futures on spot; they use DiD with treatment/control. We have a single market before/after, so we use pre/post + OLS with controls. See Literature/LITERATURE_METHODS.md.

> **Speaker notes:** Use the figure: timeline shows one market, pre and post; the OLS box shows we estimate the effect of the event (β) controlling for macro. Mention that the Bitcoin futures paper uses DiD; we cannot because we have no control group.

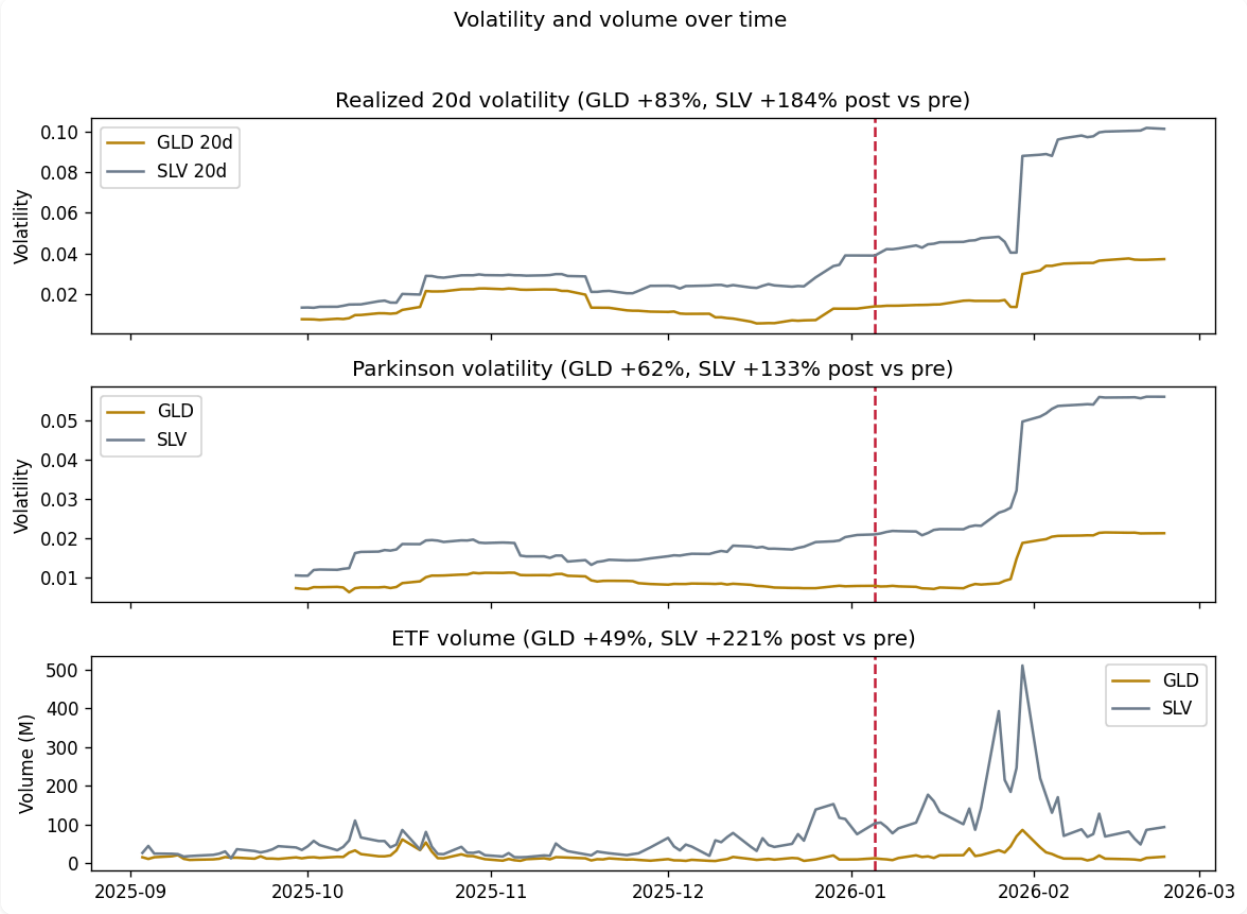
6. Cumulative returns (ETFs vs Binance)



On the chart, **higher line = higher cumulative return**: silver (SLV, XAGUSDT) outperforms gold (GLD, XAUUSDT) in this sample. **After the event**, all four series move together—that co-movement is the first visual evidence of integration between ETF and Binance markets.

> **Speaker notes:** Explain: higher line means more profitable. Silver lines sit above gold in this period. The main takeaway for our story: once Binance launches, all four lines move in sync—integration.

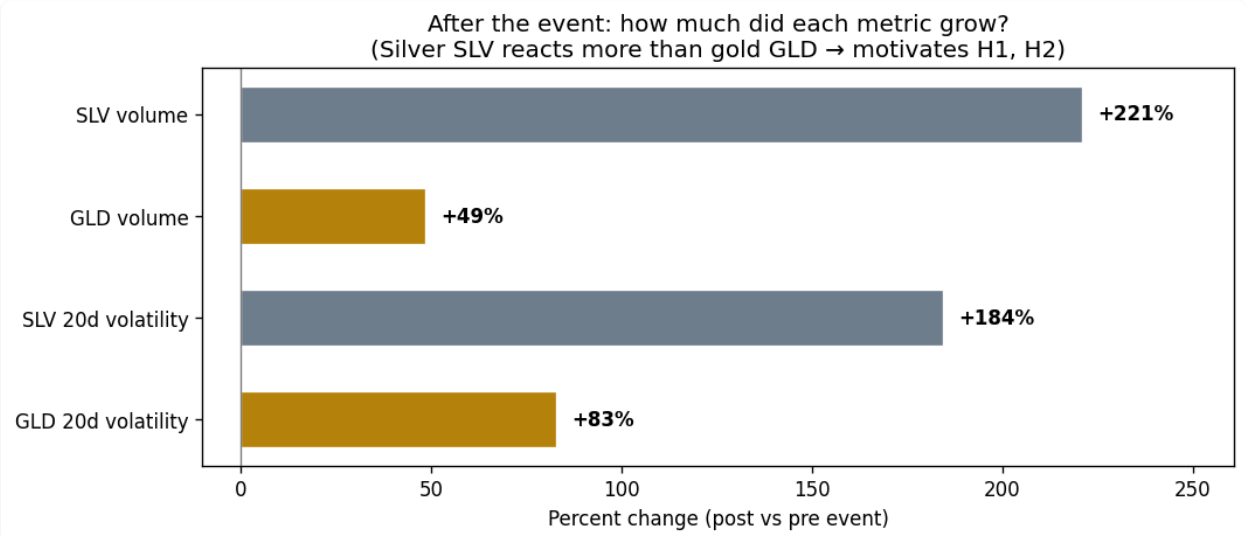
7. Volatility and volume over time



Volatility and volume dynamics around the event—basis for H1 and H2. **Silver (SLV) reacts more strongly:** post-event, both realized and Parkinson volatility, and ETF volume, rise more for SLV than for GLD.

> **Speaker notes:** Point to the red line: after the event, all three panels jump, but silver (grey line) jumps more—higher volatility and a big volume spike. This motivates testing whether the effect is significant (next slide) and robust to controls.

8. Pre vs post: how much did each metric grow?



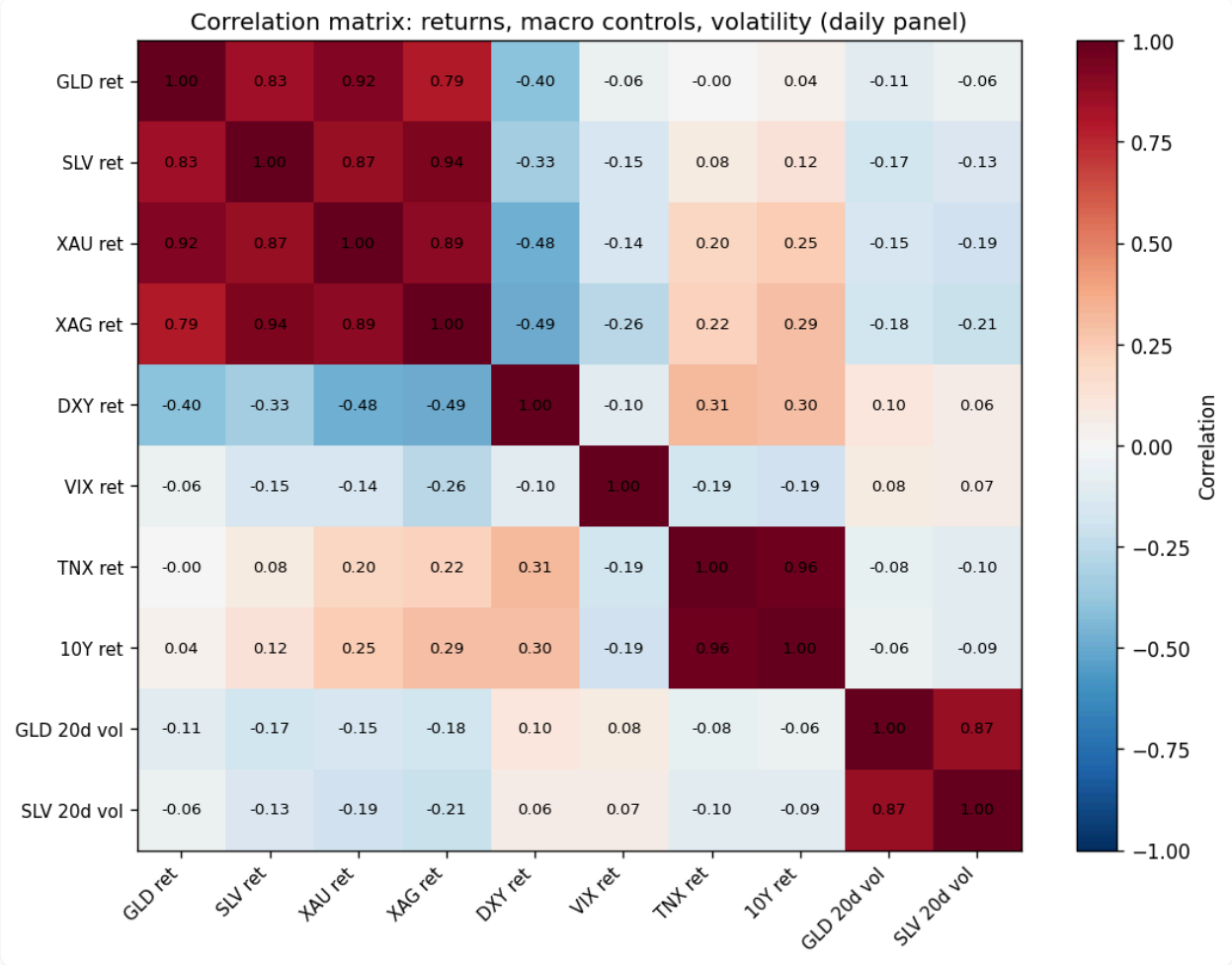
One chart, one message: after the event, volatility and volume increase (positive %); **silver (SLV) increases more than gold (GLD)**. Same units (%) make the comparison clear. Below we test whether these changes are statistically significant (t-test) and then control for macro in regressions.

T-test: For each metric we test H_0: mean(post) = mean(pre) with a **two-sample Welch t-test** (unequal variances): we take all pre-event days and all post-event days, compute the two means, and the test gives us **t** (how many standard errors the difference is) and **p** (probability of such a difference under H_0). Low p (e.g. $p < 0.05$) → reject H_0 → change is significant. Shown as `<0.001` when p is below 0.001.

Metric	Pre mean	Post mean	Pct	t	t-test p
GLD 20d vol	0.013439	0.024582	+83%	5.809	<0.001
SLV 20d vol	0.024114	0.068572	+184%	9.492	<0.001
GLD volume	15.123725	22.480038	+49%	2.377	0.0222
SLV volume	43.889215	140.791356	+221%	5.913	<0.001

> **Speaker notes:** "Horizontal bars = % change post vs pre. All four go up; the two SLV bars are longer—silver reacted more. The table shows t-test p-values: small p means the pre→post change is significant. We then control for macro in regressions."

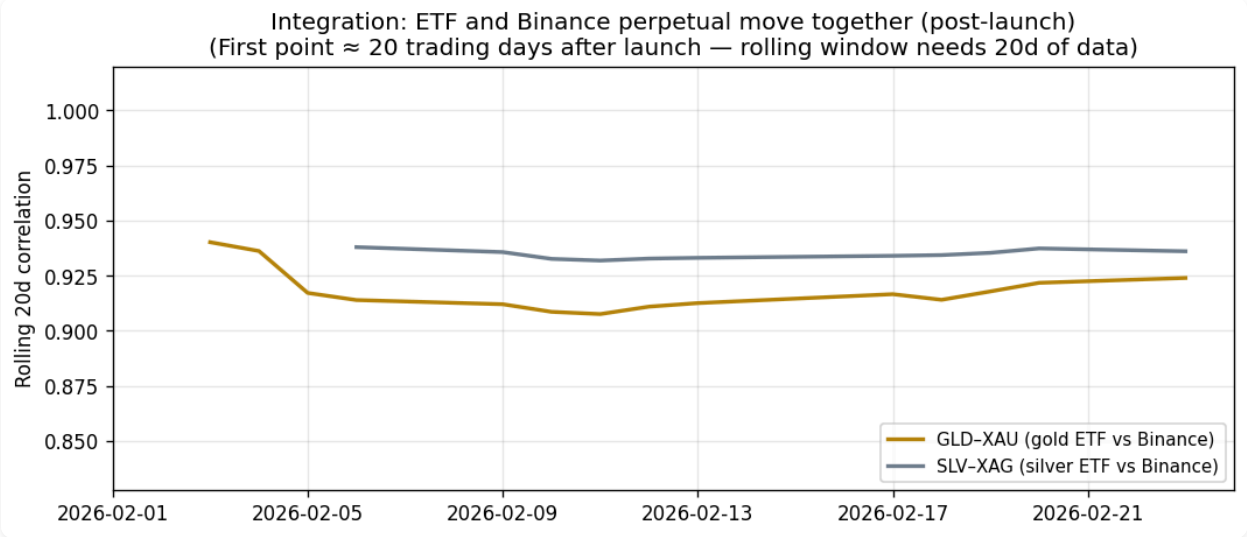
9. Correlation matrix (key variables)



What it shows: Pairwise correlations between the main variables we use: ETF returns (GLD, SLV), Binance perpetual returns (XAU, XAG), macro controls (DXY, VIX, 10Y/TNX), and realized volatility (GLD 20d, SLV 20d). Red = positive, blue = negative. High correlation between GLD and XAU (and SLV and XAG) in the post-event overlap supports integration; macro variables help us control for common factors in regressions.

> **Speaker notes:** "Correlation matrix of our key variables. ETF–Binance return pairs are highly correlated where we have data; we use DXY, VIX, rates as controls."

10. Integration: do ETF and Binance perpetual move together?



What the chart shows: Rolling 20-day correlation between ETF returns and Binance perpetual returns (GLD–XAU, SLV–XAG). We have data only after the launch; the first point appears ≈20 trading days after the event (rolling window needs 20 days). **Values near 0.9–1.0** mean the two venues move in lockstep → one price-discovery process, arbitrage and hedging link them. We cannot test "integration increased" because we have no pre-event Binance data; we describe the post-event level.

> **Speaker notes:** "Correlation is high and stable after launch—ETF and Binance perpetual are integrated. The chart starts where data exists; we don't have integration data before the event."

11. Regression results

Key result: Positive and significant coefficient on `post_event` in volatility regressions (GLD and SLV) **when we control for DXY, VIX, 10Y** (from Yahoo and FRED) → **H1 supported**. So the rise in volatility is attributed to the listing, not to general market moves. Volume higher post-event → **H2 supported** (direction). The table includes **Corr GLD–XAU** and **Corr SLV–XAG**: post-event means only (Pre = —); that is our price-integration data.

Metric	Pre mean	Post mean	Pct	t	t-test p	post coef	reg p
GLD 20d vol	0.013439	0.024582	+83%	5.809	<0.001	0.011	0.008
SLV 20d vol	0.024114	0.068572	+184%	9.492	<0.001	0.0435	0
GLD volume	15.123725	22.480038	+49%	2.377	0.0222	—	—

Metric	Pre mean	Post mean	Pct	t	t-test p	post coef	reg p
SLV volume	43.889215	140.791356	+221%	5.913	<0.001	—	—
Corr GLD–XAU	—	0.918119	—	—	—	—	—
Corr SLV–XAG	—	0.934662	—	—	—	—	—

Insight: Higher ETF volatility need not mean “chaos”—it can reflect more hedging and informed trading across the two venues.

> **Speaker notes:** Emphasize: we control for macro; the post_event effect is the causal interpretation. Mention the non-obvious reading of volatility (hedging/informed flow).

12. Conclusions

In short: The listing of Binance gold/silver perpetuals is associated with higher ETF volatility and volume (H1, H2 supported when we control for macro). The two venues move together after the launch—integration is high where we can measure it. So new derivatives on a new venue can shift incumbent markets; the story is similar to Bitcoin futures (Augustin et al.).

- **Findings:** Volatility and volume up post-event (t-tests and OLS with DXY, VIX, 10Y). Rolling correlation GLD–XAU, SLV–XAG high after launch.
- **Who cares:** ETF market-makers, arbitrageurs, regulators—two venues, one price-discovery process.
- **Limitations:** No pre-event Binance data (no “integration increased” test); no control group (no DiD); short post window—more data would help.

> **Speaker notes:** One slide: what we found, why it matters, what we couldn’t do. Thank the audience and open for questions.